

Introduction to Contract Theory and Auctions

Bayesian Auctions

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

ECON 403/503
Fall 2022

- The theory of auctions is considered to be one of the **most successful applications of game theory**, in particular of games of incomplete information.
- We will model auctions as games with incomplete information, i.e., as a Bayesian game.
- The main interest is to study the Bayesian Nash Equilibrium of the game, i.e., the equilibrium bidding strategies of the buyers and what this equilibrium strategies imply in terms of revenue for the seller.
- Moral of the theory: incomplete information matters! Most situations we encounter in daily life feature incomplete information, an auction theory is a gateway to begin to understand them (from a theoretical perspective).
- The theory of auctions is quite elegant, but is not simple.
- The mathematical challenge together with the clear applicability makes the theory of auctions **central** to modern economics.

- Under some conditions, all auction formats yield the same expected revenue for the auctioneer: this is the celebrated **Revenue Equivalence Theorem**.
- Expected revenues for the auctioneer increase the more risk averse buyers are.
- Extending the theory to arbitrary selling mechanisms, we will prove the **revelation principle**, which will allow us to consider a very simple class of selling mechanisms: **incentive-compatible direct selling mechanisms**. Again, under some conditions, a version of the revenue equivalence theorem holds for selling mechanisms. Furthermore, we will use this to show that the optimal auction in terms of revenues for the auctioneer is a **second price auction with a reserve price**.

Definition 5.1

A **sealed-bid auction** is a tuple $(N, \{T_i, v_i\}_{i \in N}, \mu, q, p)$ where:

- $N = \{1, 2, \dots, n\}$ is the set of buyers or bidders.
- T_i is the set of **types** for player i , i.e., i 's private information. We denote by $T = \prod_{i \in N} T_i$.
- μ is the **common prior** over T .
- $v_i : T \rightarrow \mathbb{R}$ is player i 's **valuation function**.
- $q : [0, \infty)^n \rightarrow \Delta(N)$ is a function associating each vector of bids $b \in [0, \infty)^n$ with a distribution according to which the buyer who wins the object is identified.
- $p : N \times [0, \infty)^n \rightarrow \mathbb{R}^n$ is a function determining the payment each buyer pays, for each vector of bids $b \in [0, \infty)^n$, depending on which buyer $i_* \in N$ is the winner.

Timeline of the auction

- A vector of types $t = (t_1, \dots, t_n)$ is realized according to μ and the i -th component t_i is disclosed to player i .
- Every buyer i submits a bid $b_i \in [0, \infty)$ depending on his type t_i .
- The buyer who wins the auctioned object, i_* , is chosen according to the distribution $q(b_1, \dots, b_n)$; the probability that buyer i wins the object is $q_i(b_1, \dots, b_n)$.
- Every buyer i pays the sum $p_i(i_*, b_1, \dots, b_n)$.
- The profit (payoff) for player i is:

$$q_i(b)v_i(t) - \sum_{i_* \in N} q_{i_*}(b)p_i(i_*; b)$$

or in words, if buyer i is the winner, she evaluates the object according to $v_i(t)$ and she pays $p_i(i_*; b)$ in any event (whether or not she is the winner), where i_* is the winning buyer.

- A **pure strategy** of buyer i in a sealed-bid auction is a *measurable* function

$$\beta_i : T_i \rightarrow [0, \infty)$$

- If buyer i uses pure strategy β_i when her type is t_i , then she bids $\beta_i(t_i)$.
- If the buyers use the strategy profile $\beta(t) = (\beta_i(t_i))_{i \in N}$, buyer's i ex-ante expected profit is:

$$\int_{t \in T} (q_i(\beta_1(t_1), \dots, \beta_n(t_n))v_i(t) - \sum_{i_* \in N} q_{i_*}(\beta(t))p_i(i_*; \beta_1(t_1), \dots, \beta_n(t_n))) d\mu(t)$$

- The interim expected profit is:

$$\int_{t_{-i} \in T_{-i}} (q_i(\beta_1(t_1), \dots, \beta_i(t_i), \dots, \beta_n(t_n))v_i(t) - \sum_{i_* \in N} q_{i_*}(\beta(t))p_i(i_*; \beta_1(t_1), \dots, \beta_n(t_n))) d\mu(t_{-i}|t_i)$$

Private Values vs. Interdependent values

- When the valuation functions $v_i : T \rightarrow \mathbb{R}$ is constant across $t_{-i} \in T_{-i}$, we say that the environment exhibits **private values**, otherwise we say that it exhibits **interdependent values**.
- Under private values, we can redefine $v_i(t_i, t_{-i}) = \hat{v}_i(t_i)$ for some $\hat{v}_i$ and all $t_{-i} \in T_{-i}$.
- With private values, types have a concrete interpretation: they define the valuation of player i for the object being sold.
- On the other hand, with interdependent values, the type of a player t_i represents much more than just a valuation, but also some information that the player might have that will influence the value of the object. That is, only by knowing the full type vector t would any player have certainty about the true value of the object.

- Consider the prior over $T: \mu(t)$
- It maybe the case that it exhibits **statistical independence**:

$$\mu(t_1, \dots, t_n) = \prod_{i \in N} \text{marg}_{T_i} \mu(t_i)$$

where $\text{marg}_{T_i} \mu(t_i) = \int_{t_{-i} \in T_{-i}} d\mu(t_i, t_{-i})$ are the marginal distributions.

- If this is the case, then we say that the model has a **statistically independent prior**, otherwise, we say that it is **correlated**.

Definition 5.2

A **independent private values sealed-bid auction** is a tuple $(N, \{\mathcal{V}_i, F_i\}_{i \in N}, q, p)$ where:

- $N = \{1, 2, \dots, n\}$ is the set of buyers or bidders.
- $\mathcal{V}_i \subset \mathbb{R}$ is the set of **possible values** for buyer i for each $i \in N$. Denote $\mathcal{V} = \prod_{i \in N} \mathcal{V}_i$.
- For each buyer $i \in N$, F_i is the cumulative distribution function over her set of values $\mathcal{V}_i$. Let $F = F_1 \times F_2 \times \dots \times F_n$.

- $q : [0, \infty)^n \rightarrow \Delta(N)$ is a function associating each vector of bids $b \in [0, \infty)^n$ with a distribution according to which the buyer who wins the object is identified.
- $p : N \times [0, \infty)^n \rightarrow \mathbb{R}^n$ is a function determining the payment each buyer pays, for each vector of bids $b \in [0, \infty)^n$, depending on which buyer $i_* \in N$ is the winner.
- For each type (value) vector $v \in \mathcal{V}$, buyer's i profit is given by:

$$u_i(b, v_i) = q_i(b)v_i - \sum_{i_* \in N} q_{i_*}(b)p_i(i_*; b)$$

Independent private values sealed-bid auction: payoffs

- Let $\beta_i : \mathcal{V}_i \rightarrow [0, \infty)$ be a pure-strategy for each player i . Let $\beta(v) = (\beta_1(v_1), \dots, \beta_n(v_n))$ be a strategy profile.
- The ex-ante expected payoff to player i is:

$$U_i(\beta) = \int_{v \in \mathcal{V}} u_i(\beta_1(v_1), \dots, \beta_n(v_n)) dF(v)$$

- Let $\beta_{-i}(v_{-i}) = (\beta_j(v_j))_{j \neq i}$, i.e., the vector of pure strategies of all players except i .
- The interim expected payoff to player i when her type is $V_i = x$ is:

$$U_i(\beta; x) = \int_{v_{-i} \in \mathcal{V}_{-i}} u_i(\beta_i(x), \beta_{-i}(v_{-i})) dF_{-i}(v_{-i})$$

where $\mathcal{V}_{-i} = \prod_{j \neq i} \mathcal{V}_j$ is the space of buyer values other than i , and $F_{-i} = \prod_{j \neq i} F_j$ is the cumulative joint distribution function of the random variable $V_{-i} = \prod_{i \in N} V_j$.

- Notice that the interim expected profit of buyer i when her value is v_i depends on β_i , buyer i 's strategy, only via $\beta_i(v_i)$, which is the bid of buyer i with private value v_i .
- Denote by $U_i(b_i, \beta_{-i}; v_i)$ the expected profit of buyer i with private value v_i when she submits a bid b_i and the other buyers use the profile β_{-i} .

Definition 5.3

A strategy profile β^* is an equilibrium (Bayesian-Nash eq.) if for every buyer $i \in N$, and every private value $v_i \in \mathcal{V}_i$:

$$U_i(\beta^*; v_i) \geq U_i(b_i, \beta_{-i}^*; v_i), \quad \forall b_i \in [0, \infty).$$

Equilibrium in the sealed-bid second price auction

- Let $N(b) = \{i \in N : b_i = \max_{j \in N} b_j\}$.
- The function $q(b) \in \Delta(N)$ of the second-price auction is:

$$q_i(b) = \begin{cases} 0 & i \notin N(b) \\ \frac{1}{|N(b)|} & i \in N(b) \end{cases}$$

- Let i_* denote the buyer who wins the auctioned object, then the payment rule $p(i_*; b)$ is:

$$p_i(i_*; b) = \begin{cases} 0 & i \neq i_* \\ \max_{j \neq i} b_j & i = i_* \end{cases}$$

- Notice that $u_i(b, v_i)$ becomes:

$$u_i(b, v_i) = \begin{cases} 0 & b_i < \max_{j \in N} b_j \\ \frac{v_i - \max_{j \neq i} b_j}{|N(b)|} & b_i = \max_{j \in N} b_j \end{cases}$$

Theorem 5.4

In a sealed-bid second price auction, the strategy of buyer i in which he bids his private value (ex-post) weakly dominates her other strategies.

Theorem 5.5

In a sealed-bid second price auction (with private values), the strategy profile in which every buyer's bid equals her private value is an equilibrium.

In an open-bid ascending auction (like the English auction), the strategy profile in which every buyer lowers her hand when the declared price equals her private value is an equilibrium